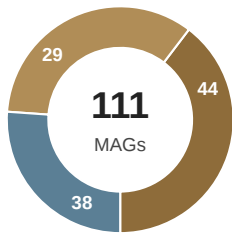


Cattle manure-soil microcosm MAGs yield two convergent lineages with a stably retained *ure-cah* module

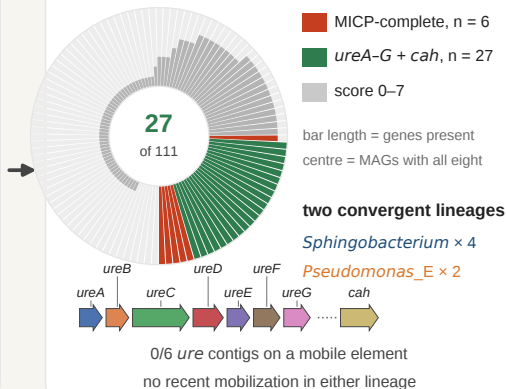
a genomic chassis for alkali-tolerant microbially induced carbonate precipitation (MICP)

111 enrichment-culture MAGs



dairy-cow manure · soil microcosms · day 2–28

MICP module score across the panel



What the six MICP-complete MAGs share

42/42

UreC active-site residues
identical to *S. pasteurii*

11.7×

Mrp Na⁺/H⁺ antiporter prevalence
the alkali-tolerance signature

$\omega \leq 0.09$

codeml M0 on *ureA/B/C/G*
strong purifying selection

S13 · S16

novel *Sphingobacterium*
ANI 94.6 / 93.8 % to RefSeq

urease fold kept 6/6 · ESMFold TM > 0.5

waste-coupled MICP

